

Scalable Parallel-in-Time Integration for Particle Production in Analog Cosmologies

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Abstract

Something abstract-like goes here

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1 Introduction

Directly measuring the properties of the universe just after the Big Bang is impossible, as that was almost 14 billion years ago. Even *indirectly* measuring these properties is extremely difficult via traditional means. During these brief moments just after the Big Bang, the universe expanded rapidly in a particular way. During this expansion there were particles popping in and out of existence, each with its own dynamics (e.g. position and momentum).

What is less difficult is cooling down gases to near absolute-zero (about a billion times colder than empty space). Gases made of certain atoms or molecules have properties that can be changed to almost any value we want. Because of this, we can turn our knobs in the lab to make the gas behave in a way that matches a certain mathematical model.

For some types of gases, we can choose how strongly the particles in the gas interact with each other. It turns out that we can choose a certain interaction strength so that the mathematical model that describes how the gas behaves **exactly** matches the mathematical model of how particles are produced in the universe in the moments just after the Big Bang! When this happens, we can call this ultra-cold gas an “analog universe”.

Because of this mathematical equality, we can effectively observe the particles in the moments just after the Big Bang, but in the lab. Moreover, these observations can be made using a tried-and-true system that has been developed and used over the past thirty years[1].

Studies have investigated gases with

- phonons with a temporal frequency that is nonlinearly related to the spatial frequency (dispersion relation)[3, 25, 32]
- analog massive particles[18, 31, 29]
- analog particles with non-zero spin[19, 27, 26, 21]
- analog universes with different expansion behavior[13, 24, 22, 2]
- analog universes with non-zero background velocity[30, 7, 8, 6, 23]
- promising experimental realizations [20, 5, 2, 28]

Computationally, simulations have had to investigate analog universes of reduced dimension due to the unreasonable time it takes to simulate full-dimensional systems without using high-performance computing. These simulations have been deemed accurate enough as it is proposed that a full-dimensional simulation would yield qualitatively similar results. In order to be confident in the level of accuracy of the reduced-dimensional simulations, those results need to be compared to those of a full-dimensional simulation. This will not only allow a quantification of the error in the reduced-dimensional simulation, but also a measure of the error-to-resource efficiency of a full-dimensional simulation.

Future insight into these analog systems and the fundamental properties of the early universe require computational support. A readily-usable computational model will not only allow theoretical investigations to make predictions, but also allow experimental research to have a guide on where to go next and have something with which to compare. The availability of this work is paramount to more efficient and more physically-accurate insight into our universe.

I choose a gas with linear dispersion, analog massless and spin-0 particles, an analog universe undergoing a de Sitter expansion with zero background velocity. The de Sitter spacetime is chosen because of its significance to modern cosmology and previously predicted particle production (as done by Hawking)[12, 11, 16]. Particle production will be calculated with standard methods[4, 9]. The numerical parameters chosen in this study also follow from previous studies[10]. Computational implementations will be done using high-performance methods.

2 Methods

The methods used in this investigation fall into two different categories: the physical theory of the production of phononic quasiparticles in analog inflationary cosmologies arising from the dynamics of Bose-Einstein condensates, and the computational implementations of high-performance methods such as adaptive integration, finite-differences on non-uniform grids, discretizing differential operators, and the Fast-Fourier Transform.

2.1 Physical Theory

2.1.1 BEC Analogs of FLRW Cosmologies

The gas as is outlined in section 1 is a ground-state Bose-Einstein condensate (BEC) without thermal or quantum fluctuations. The dynamics of the BEC are described by the Gross-Pitaevskii equation (GPE) under a Bogoliubov mean-field approximation (also known as the *nonlinear Schrödinger equation*),

$$i\hbar \partial_t \psi(t, \mathbf{x}) = \left[-\frac{\hbar^2}{2m} \nabla^2 + V_{\text{ext}}(x) + U |\psi(t, \mathbf{x})|^2 \right] \psi(t, \mathbf{x}). \quad (1)$$

The wave function ψ can be expanded using a linearized Madelung density-phase representation

$$\psi(t, \mathbf{x}) = \sqrt{n_0 + \Delta n(t, \mathbf{x})} e^{i(\theta_0 + \Delta \theta(t, \mathbf{x}))}, \quad (2)$$

where $n_0 + \Delta n(t, \mathbf{x})$ and $\theta_0 + \Delta \theta(t, \mathbf{x})$ are the linearized forms of the real density and phase fields, respectively. Furthermore, $\Delta n(t, \mathbf{x})$ and $\Delta \theta(t, \mathbf{x})$ are the density and phase perturbations, respectively.

With the Madelung representation defined in equation (2), the GPE in equation (1) becomes[14]

$$\frac{1}{\sqrt{-g}} \partial_\mu [\sqrt{-g} g^{\mu\nu} \partial_\nu \Delta \theta] = 0, \quad (3)$$

where

$$g_{\mu\nu} = \left(\frac{n_0}{c} \right)^{2/(d-1)} \begin{bmatrix} -c^2 & \vdots & 0 \\ \cdots & & \cdots \\ 0 & \vdots & \delta_{ij} \end{bmatrix}. \quad (4)$$

Equation (4) can be interpreted as the covariant metric tensor (with determinant g) describing an analog, spatially flat, Friedmann–Lemaître–Robertson–Walker universe, c is the speed of sound in the condensate, and d is the number of spatial dimensions. Equation (3) describes both the phase-perturbations $\Delta\theta$ of oscillations with low spatial frequency (i.e. low momentum phonons) in the BEC and also the dynamics of a quantum field that produces massless, spin-0 particles.

2.1.2 Variable Speed of Sound and Inflation

The time dependence of the system is completely captured in the speed of sound by

$$c(t)^2 = \frac{U(t)n_0}{m} = \frac{4\pi\hbar^2}{m^2} n_0 \ell(t), \quad (5)$$

with atoms of mass m , scattering length $\ell(t)$, and number density n_0 . With the dimensionless scaling function $a(t)$, the interaction strength $U(t)$ (or equivalently the scattering length) has time dependence defined by

$$U(t) \equiv U_0 a(t), \quad (6)$$

where $U_0 = U(t_0)$ for an initial time t_0 . Then equation (5) becomes

$$c(t) = c_0 \sqrt{a(t)}. \quad (7)$$

This time dependence of the speed of sound and interaction strength allows us to draw another analogy. If the speed of sound decreases with time, it would appear as if the space between the source and the observer were increasing. Similarly, if the speed of sound increases with time, it would appear as if the space between the source and observer is decreasing. With this in mind, if the interaction strength between atoms $U(t)$ decreases with t , the analog universe is expanding, and if $U(t)$ increases, the analog universe is contracting.

2.1.3 The Field Equation

With the scaling function, equation (3) becomes¹

$$\text{2D: } \partial_t^2 \Delta\theta - \frac{3}{2} \frac{\dot{a}(t)}{a(t)} \partial_t \Delta\theta - c_0^2 a(t) \nabla^2 \Delta\theta = 0 \quad (8a)$$

$$\text{3D: } \partial_t^2 \Delta\theta - \frac{\dot{a}(t)}{a(t)} \partial_t \Delta\theta - c_0^2 a(t) \nabla^2 \Delta\theta = 0 \quad (8b)$$

These equations only differ in the constant coefficient to the first derivative i.e. the dissipative term. If we consider the *conformal time* η defined by $d\eta = \sqrt{a(t)} dt$, the two-dimensional field equation (eq 8a) becomes

$$\partial_\eta^2 \Delta\theta - \frac{1}{2} \frac{\dot{a}(\eta)}{a(\eta)} \partial_\eta \Delta\theta - c_0^2 \nabla^2 \Delta\theta = 0. \quad (9)$$

¹ $\dot{a}(t)/a(t)$ is the Hubble parameter for an expanding universe with scaling parameter $a(t)$.

Equation (9) governs the dynamics of oscillations in the BEC for a time-dependent speed of sound, and the behavior of the massless, spin-0 particles in an expanding universe.

2.1.4 Phononic & Free-Particle Modes

The phase $\Delta\theta(t, \mathbf{x})$ and density $\Delta n(t, \mathbf{x})$ perturbation fields can be expanded in a Fourier plane-wave amplitude representation as

$$\Delta\theta(t, \mathbf{x}) = \frac{1}{\sqrt{V}} \sum_{\mathbf{k}} e^{i\mathbf{k}\cdot\mathbf{x}} \tilde{\theta}_{\mathbf{k}}(t) \hat{b}_{\mathbf{k}}(t) + e^{-i\mathbf{k}\cdot\mathbf{x}} \tilde{\theta}_{\mathbf{k}}^*(t) \hat{b}_{\mathbf{k}}^\dagger(t) \quad (10a)$$

$$\Delta n(t, \mathbf{x}) = \frac{1}{\sqrt{V}} \sum_{\mathbf{k}} e^{i\mathbf{k}\cdot\mathbf{x}} \tilde{n}_{\mathbf{k}}(t) \hat{b}_{\mathbf{k}}(t) + e^{-i\mathbf{k}\cdot\mathbf{x}} \tilde{n}_{\mathbf{k}}^*(t) \hat{b}_{\mathbf{k}}^\dagger(t) \quad (10b)$$

where $\mathbf{k}$ is the wave-vector of the phase and density perturbations, $\tilde{\theta}$ is the discrete Fourier Transform of the phase perturbation, $\hat{b}$ and $\hat{b}^\dagger$ are the annihilation and creation operators, respectively. From this, there arises two domains to consider²: phononic quasiparticles with linear dispersion, and free-particle-like quasiparticles with quadratic dispersion.

The boundary between phononic and free-particle-like quasiparticles is described by the critical wave-number k_c . The critical wave-number k_c is defined[14] in terms of the healing length of the BEC ξ , where

$$\xi(t) = \frac{\hbar}{\sqrt{2mc}} = \xi_0/\sqrt{a(t)}, \quad (11)$$

and

$$k_c(t) = 1/\xi(t) = \sqrt{a(t)}/\xi_0. \quad (12)$$

Wave-vectors $\mathbf{k}$ with wave-number $k = ||\mathbf{k}||$ such that $k \ll k_c$ are called *phononic*³. Likewise, those with $k \gg k_c$ are called *free-particle-like*⁴.

For phononic quasiparticles, the analog between a BEC and an inflationary cosmology is exact. Including free-particle-like quasiparticles forces the analog to include ‘trans-Planckian’ effects and analog Lorentz violation[14]. For these reasons, this investigation focuses only on phononic quasiparticles; namely wave-vectors $\mathbf{k}$ such that $||\mathbf{k}|| \ll k_c$. Additionally, the free-particle-like regime has nonlinear dispersion which significantly increases the complexity of the field equation[14].

Only considering phononic modes, the two-dimensional field equation (eq. 9) then becomes

$$\partial_\eta^2 \tilde{\theta}_{\mathbf{k}} - \frac{1}{2} \frac{\dot{a}(\eta)}{a(\eta)} \partial_\eta \tilde{\theta}_{\mathbf{k}} - c_0^2 k^2 \tilde{\theta}_{\mathbf{k}} = 0. \quad (13)$$

²These domains can also be described as “acoustic” and “Bogoliubov”, respectively.

³Phononic quasiparticles have linear dispersion $\omega = ck$ characterized by neglecting quantum pressure.

⁴Free-particle-like quasiparticles have quadratic dispersion characterized by $\omega_k(t)^2 = \frac{k^2}{2m} \left(\frac{\hbar^2 k^2}{2m} + 2U(t)n_0 \right)$ characterized by including quantum-pressure

For a de Sitter universe, the scaling function is defined such that $a(t) = \exp(-t/t_s)$. Therefore, the field equation for a two-dimensional de Sitter universe is

$$\partial_\eta^2 \tilde{\theta}_{\mathbf{k}} - \frac{1}{\eta} \partial_\eta \tilde{\theta}_{\mathbf{k}} + c_0^2 k^2 \tilde{\theta}_{\mathbf{k}} = 0, \quad (14)$$

2.1.5 Initial Conditions

As the phase and density perturbations need to be continuous at $t = 0$ when the expansion begins, the initial conditions i.e. spacetime boundary conditions are defined by this physical conservation. Therefore

$$\tilde{\theta}_{\mathbf{k}}(t = 0) = \tilde{\theta}_{\mathbf{k}0} \quad (15a)$$

$$\partial_t \tilde{\theta}_{\mathbf{k}}(t = 0) = -\frac{U}{\hbar} \tilde{n}_{\mathbf{k}0} \quad (15b)$$

where the initial derivative of the phase perturbation's Fourier amplitude (eq. 15b) is determined by neglecting the quantum pressure[14].

2.1.6 Particle Production

The number of particles $N_k(t)$ produced at time t with wave-vector $\mathbf{k}$ is described by the equation

$$N_k(t) = \left| u_k^{\text{out}*}(t) v_k^{\text{exp}*}(t) - v_k^{\text{out}*}(t) u_k^{\text{exp}*}(t) \right|^2 \quad (16)$$

where

$$u_k(t) = \frac{1}{2\sqrt{n_0}} \tilde{n}_k(t) + i\sqrt{n_0} \tilde{\theta}_k(t) \quad (17a)$$

$$v_k(t) = \frac{1}{2\sqrt{n_0}} \tilde{n}_k(t) - i\sqrt{n_0} \tilde{\theta}_k(t), \quad (17b)$$

“out” and “exp” corresponding to regions of spacetime outside of and inside the expansion of the universe.

To summarize, to calculate the number of particles produced at position $\mathbf{x}$ and time t with wave-vector $\mathbf{k}$ looks like the following:

1. Solve the field equation (13) for $\tilde{\theta}_{\mathbf{k}}(t)$
2. Differentiate $\tilde{\theta}_{\mathbf{k}}(t)$ to get $\tilde{n}_{\mathbf{k}}(t)$ as in equation (15b)
3. Combine $\tilde{\theta}_{\mathbf{k}}(t)$ and $\tilde{n}_{\mathbf{k}}(t)$ as in equations (17)
4. Calculate the number of particles at time t with wave-vector $\mathbf{k}$ as in equation (16)

2.2 Computational Implementation

2.2.1 Parallel-in-Time Integration & The Parareal Algorithm

- The Parareal algorithm is used to solve initial value differential equations in parallel.
- It uses a computationally cheap “coarse” propagator $\mathcal{C}$ and a more computationally expensive “fine” propagator $\mathcal{F}$.
- It is an iterative method along the likes of a predictor-corrector, or shooting method.
- It converges linearly or superlinearly in iterations.
- The basic Parareal algorithm doesn’t conserve energy no matter if the coarse and fine propagators are symplectic
- Because there are several drawbacks to the basic Parareal algorithm, variants have been developed to address some of these issues. For example, ParaExp uses exponential matrix integration on linear problems, or so-called “Krylov-Enhanced” methods use geometric methods to make the base algorithm more accurate and stable.

Algorithm 1: The Parareal Algorithm

Input: Coarse Propagator $\mathcal{C}$, Fine Propagator $\mathcal{F}$, Initial Condition u_0 , Discretized Time Domain

$T = \{T_0, T_1, \dots, T_N\}$

Output: Solution $\{u_0, u_1, \dots, u_N\}$

// INITIALIZATION

1 Use the coarse propagator $\mathcal{C}$ and initial value u_0 to get data points $\{u_0, u_1, \dots, u_N\}$ on the discretized time domain T

// BEGIN PARALLEL

// parallelize over the temporal subdomains

2 **for** $n = 0, 1, \dots, N - 1$ **do**

3 Use the fine propagator $\mathcal{F}$ and psuedo-initial value u_n to get the data point $u_{n+1}^{\mathcal{F}}$

4 Use the coarse propagator $\mathcal{C}$ and psuedo-initial value u_n to get the data point $u_{n+1}^{\mathcal{C}}$

5 $corrector_{n+1} \leftarrow u_{n+1}^{\mathcal{F}} - u_{n+1}^{\mathcal{C}}$

2.2.2 Supercomputing with Julia

- Distributed computing across nodes with `Distributed.jl`
- GPU computing with CUDA and multiple cards with `CUDA.jl`

3 Results

4 Discussion

4.1 Physical Implications

4.2 Numerical Analysis

4.2.1 Error

4.2.2 Convergence

4.2.3 Stability

4.3 Algorithm Analysis

4.3.1 Time Complexity

4.3.2 Space Complexity

4.4 Benchmarks

5 Conclusion

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